

Guide to Typst

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1 | Introduction

The goal of this document is to have the most common used elements for the markup language **typst** readily available. A detailed documentation can be found on their website: <https://typst.app/docs> It is to note that these are **my** most common used elements. For some elements custom templates are needed:

- [codelst](#)
- [glossarium](#)
- [tablex](#) (*for legacy tables only*)
- all files and folders in the **/00-templates/** folder such as
 - **icons/*.svg**
 - **scripts/*.bash**
 - **syntax/*.sublime-syntax**
 - **boxes.typ**
 - **constants.typ**
 - **helpers.typ**
 - **il8n.json**
 - **items.typ**
 - **karnaugh.typ**
 - **page-*.typ**
 - **sections.typ**
 - **template-***

2 | Installation

2.1 With cargo

If you use already the **rust** programming language then you can use rust to install the latest toolchain.

```
# install rust and cargo
curl https://sh.rustup.rs -sSf | sh

# install typst
cargo install --git https://github.com/typst/typst
```

2.2 MacOS

On MacOS you can use **homebrew**

```
# install homebrew
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"

# install typst
brew install typst
```

2.3 Linux

In Linux you can use the commonly available package manager

```
brew install typst
pacman -S typst
xbps-install typst
sudo apt-get install typst
```

2.4 Windows

On Windows you can use **chocolatey**. See: <https://chocolatey.org/install>

```
# install chocolatey
# ensure to use a administrative powershell
Set-ExecutionPolicy
Bypass -Scope Process -Force; [System.Net.ServicePointManager]::SecurityProtocol
= [System.Net.ServicePointManager]::SecurityProtocol -bor 3072; iex ((New-Object
System.Net.WebClient).DownloadString('https://community.chocolatey.org/install.ps
1'))

# install typst
choco install typst
```

3 | Formatting

3.1 Markup

Name	Example	Raw
Singleline Comment		//
Multiline Comment		/* */
Paragraph break		blankline
Line break		\
bold	bold	*bold*
italic	<i>italic</i>	_italic_
monospaced	monospaced	`monospaced`
math	$x = 1$	\$x=1\$
lowercase	lower	#lower("LoWeR")
uppercase	UPPER	#upper("UpPeR")
smallcaps	SMALLCAPS	#smallcaps("SmallCaps")
smartquote	« test »	#smartquote()test#smartquote()
overline	<u>overline</u>	#overline("overline")
underline	<u>underline</u>	#underline("underline")
strike	strike	#strike("strike")
sub	Text _{sub}	Text#sub("sub")
super	Text ^{super}	Text#super("super")
Label		<label>
Reference		@label

3.2 Page Formatting

```
#pagebreak() // pagebreak
#parbreak() // parbreak
\ // linebreak
```

3.3 Space

A

B

A #h(5cm) B,

C

C #v(0.2cm) D

D

3.4 Text Formatting

For the custom textsizes and colors you need to import:

```
#import "/01-tail/constants.typ": *
```

Name	Example	Raw
Sizes	8pt text tiny text	<code>text(8pt, "8pt text") text(tiny "tiny text")</code>
	9pt text smaller text	<code>text(9pt, "9pt text") text(smaller "smaller text")</code>
	10pt text small text	<code>text(10pt, "10pt text") text(small "small text")</code>
	11pt text normal text	<code>text(11pt, "11pt text") text(normal "normal text")</code>
	14pt text large text	<code>text(14pt, "14pt text") text(large "large text")</code>
	16pt text larger text	<code>text(16pt, "16pt text") text(larger "larger text")</code>
	24pt text huge text	<code>text(24pt, "24pt text") text(huge "huge text")</code>
	36pt text huger text	<code>text(36pt, "36pt text") text(huger "huger text")</code>
	Fira Sans	<code>text(font:"Fira Sans", "Fira Sans")</code>
	Fira Mono	<code>text(font:"Fira Mono", "Fira Mono")</code>
Types	Source Sans Pro	<code>text(font:"Source Sans Pro", "Source Sans Pro")</code>
	Arial	<code>text(font:"Arial", "Arial")</code>
	Times New Roman	<code>text(font:"Times New Roman", "Times New Roman")</code>
Alignment	start	<code>align(start){start}</code>
	end	<code>align(end){end}</code>
	left	<code>align(left){left}</code>
	center	<code>align(center){center}</code>

Name	Example	Raw
	right	<code>align(right){right}</code>
	top	<code>align(top){top}</code>
	horizon	<code>align(horizon){horizon}</code>
	bottom	<code>align(bottom){bottom}</code>
	center + horizon	<code>align(center + horizon){center + horizon}</code>
Colors	black	<code>#text(fill:black)[black]</code>
	red	<code>#text(fill:red)[red]</code>
	green	<code>#text(fill:green)[green]</code>
	blue	<code>#text(fill:blue)[blue]</code>
	purple	<code>#text(fill:purple)[purple]</code>
	gray-80	<code>#text(fill:gray-80)[gray-80]</code>
	gray-70	<code>#text(fill:gray-70)[gray-70]</code>
	gray-60	<code>#text(fill:gray-60)[gray-60]</code>
	gray-50	<code>#text(fill:gray-50)[gray-50]</code>
	gray-40	<code>#text(fill:gray-40)[gray-40]</code>
	gray-30	<code>#text(fill:gray-30)[gray-30]</code>
	gray-20	<code>#text(fill:gray-20)[gray-20]</code>
	gray-10	<code>#text(fill:gray-10)[gray-10]</code>
	hei-orange	<code>#text(fill:hei-orange)[hei-orange]</code>
	hei-blue	<code>#text(fill:hei-blue)[hei-blue]</code>
	hei-pink	<code>#text(fill:hei-pink)[hei-pink]</code>
	hei-yellow	<code>#text(fill:hei-yellow)[hei-yellow]</code>
	hei-green	<code>#text(fill:hei-green)[hei-green]</code>
	spl-green	<code>#text(fill:spl-green)[spl-green]</code>
	spl-blue	<code>#text(fill:spl-blue)[spl-blue]</code>
	spl-pink	<code>#text(fill:spl-pink)[spl-green]</code>
	color-info	<code>#text(fill:color-info)[color-info]</code>
	color-idea	<code>#text(fill:color-idea)[color-idea]</code>
	color-warning	<code>#text(fill:color-warning)[color-warning]</code>
	color-important	<code>#text(fill:color-important)[color-important]</code>
	color-fire	<code>#text(fill:color-fire)[color-fire]</code>
	color-rocket	<code>#text(fill:color-rocket)[color-rocket]</code>
	color-todo	<code>#text(fill:color-todo)[color-todo]</code>

Name	Example	Raw
	<code>code-bg</code>	<code>#text(fill:code-bg)[code-bg]</code>
	<code>code-border</code>	<code>#text(fill:code-border)[code-border]</code>

4 | Elements

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4.1 Headings

```
= Heading 1
== Heading 1.1
=== Heading 1.1.1
==== Heading 1.1.1.1
...
```

4.2 Lists

- First
- Second
- Third

```
- First
- Second
- Third
```

- First
 - Second
 - Third

```
- First
- Second
- Third
```

- First
- Second
- Third

```
- First
- Second
- Third
```

- First
- Second
- Third

```
list(
  [First],
  [Second],
  [Third],
)
```

1. First
 1. Second
2. Third

```
+ First
+ Second
+ Third
Text
4. Fourth
+ Fifth
```

Text

4. Fourth
5. Fifth

```
+ First
  #set enum(numbering: "a")
+ Second
+ Third
Text
4. Fourth
+ Fifth
```

1. First
 - a) Second
2. Third

Text

4. Fourth
5. Fifth

4.3 Custom Lists

```
#import "/00-templates/items.typ": *
```

- ☒ item-list
- ☒ item-checkbadge
- ☐ item-circle
- ☐ item-square
- ☒ item-checkcircle
- ☒ item-checksquare
- ✓ item-check
- ☐ item-file
- ☐ item-folder
- ☒ item-xcircle
- ☒ item-xs-
- quare
- × item-x

```
#item-list()[item-list]
#item-checkbadge()[item-checkbadge]
#item-circle()[item-circle]
#item-square()[item-square]
#item-checkcircle()[item-checkcircle]
#item-checksquare()[item-checksquare]
#item-check()[item-check]
#item-file()[item-file]
#item-folder()[item-folder]
#item-xcircle()[item-xcircle]
#item-xsquare()[item-xsquare]
#item-x()[item-x]
```

4.4 Minitoc

The **minitoc** is specific for this template and allows to show a TOC between two labels. The **minitoc** is also used within the **#add-chapter** function if **after** and **before** labels are defined.

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```
#minitoc(after: <sec:elem>, before: <sec:ref>)
```

```
#add-chapter(  
  "/02-main/03-elements.typ",  
  after: <sec:elem>,  
  before: <sec:ref>,  
)  
// or  
#add-chapter(  
  after: <sec:elem>,  
  before: <sec:ref>,  
) [  
  Content of the Chapter  
]
```

4.5 Images

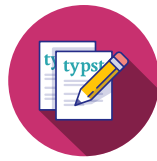
4.5.1 Alignment

left



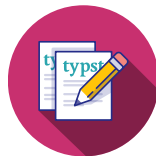
```
#image("/04-resources/icon.svg",  
width: 2cm)
```

center



```
#align(center,
  image("/04-resources/icon.svg",
    width: 2cm)
)
```

right



```
#align(right,
  image("/04-resources/icon.svg",
    width: 2cm)
)
```

4.5.2 Caption



Figure 1 - One image one caption

```
#figure(
  image("/04-resources/icon.svg",
    width: 2cm),
  caption: [One image one caption]
) <fig:icon>
```

4.5.3 Cluster

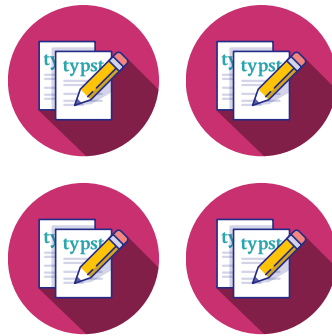
Two images one caption



Table 1 - Multiple images **one** caption

```
#figure(
  table(
    columns: 2,
    stroke: none,
    align: center + horizon,
    image(icon, width: 2cm), image(icon, width: 2cm)
  ),
  caption: [Multiple images *one* caption]
)
```

Four images one caption

Table 2 - Multiple images **one** caption

```
#figure(
  table(
    columns: 2,
    stroke: none,
    align: center + horizon,
    image(icon, width: 2cm), image(icon, width: 2cm),
    image(icon, width: 2cm), image(icon, width: 2cm),
  ),
  caption: [Multiple images *one* caption]
)
```

Two images two caption



Figure 2 - Caption left image Figure 3 - Caption right image

```
#align(center,
  table(
    columns: 2,
    stroke: none,
    align: center + horizon,
    figure(image(icon, width: 2cm), caption: [Caption left image]), figure(image(icon,
    width: 2cm), caption: [Caption right image]),
  ))
```

Four images four caption



Figure 4 - Caption topleft image

Figure 5 - Caption topright image

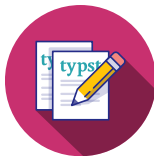


Figure 6 - Caption bottomleft image Figure 7 - Caption bottomright image

```
#align(center,
  table(
    columns: 2,
    stroke: none,
    align: center + horizon,
    figure(image(icon, width: 2cm), caption: [Caption topleft image]),
    figure(image(icon, width: 2cm), caption: [Caption topright image]),
    figure(image(icon, width: 2cm), caption: [Caption bottomleft image]),
    figure(image(icon, width: 2cm), caption: [Caption bottomright image]),
  ))
```

4.6 Tables

For new table use the integrated **#table** command for legacy the **tablex** plugin is also imported.

```
#import "@preview/tablex:0.0.9" : *
// or
#import "/00-templates/helpers.typ" : *
```

Tables with and without caption

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

Table 3 - Table caption

```
table(
  columns: 3,
  align: center + horizon,
  table.header([], [*Col1*] ,
[*Col2*]),),
[*Row1*], "cell-0-0", "cell-1-0",
[*Row2*], "cell-0-1", "cell-1-1",
)
```

```
figure(
  table(
    columns: 3,
    align: center + horizon,
    table.header([], [*Col1*] ,
[*Col2*]),),
    [*Row1*], "cell-0-0", "cell-1-0",
    [*Row2*], "cell-0-1", "cell-1-1",
  ),
  kind: table,
  caption: [Table Caption]
)
```

Tables with cell spans

	Col1	Col2
Row1	cell-0	cell-1-0
Row2		cell-1-1

```
table(  
  columns: 3,  
  align: center + horizon,  
  table.header([], [*Col1*] ,  
[*Col2*]),),  
  [*Row1*], table.cell(rowspan: 2)  
[cell-0], "cell-1-0",  
                                     [*Row2*],  
"cell-1-1",  
)
```

	Col1	Col2
Row1	cell-0	
Row2	cell-0-1	cell-1-1

```
table(  
  columns: 3,  
  align: center + horizon,  
  table.header([], [*Col1*] ,  
[*Col2*]),),  
  [*Row1*], table.cell(colspan: 2)  
[cell-0],  
  [*Row2*], "cell-0-1", "cell-1-1",  
)
```

Table Design

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

```
table(  
  columns: 3,  
  align: center + horizon,  
  table.header([], [*Col1*] ,  
[*Col2*]),),  
  [*Row1*], "cell-0-0", "cell-1-0",  
  [*Row2*], "cell-0-1", "cell-1-1",  
)
```

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

```
table(  
  columns: 3,  
  align: center + horizon,  
  table.header([], [*Col1*] ,  
[*Col2*]),),  
  [*Row1*], "cell-0-0", "cell-1-0",  
  [*Row2*], "cell-0-1", "cell-1-1",  
)
```

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

```
table(  
  columns: 3,  
  align: center + horizon,  
  table.vline(x:0, stroke: none),  
table.vline(x:1 , stroke: blue),  
table.vline(x:2),  
  table.header([], [*Col1*] ,  
[*Col2*], table.hline(stroke: red)),  
  [*Row1*], "cell-0-0", "cell-1-0",  
table.hline(),  
  [*Row2*], "cell-0-1", "cell-1-1",  
)
```

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

```
table(  
  columns: 3,  
  align: center + horizon,  
  stroke: (x:none),  
  table.header([], [*Col1*] ,  
[*Col2*]),),  
  [*Row1*], "cell-0-0", "cell-1-0",  
  [*Row2*], "cell-0-1", "cell-1-1",  
),
```


	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

```
table(  
  columns: 3,  
  align: center + horizon,  
  stroke: (y:none),  
  table.header([], [*Col1*] ,  
[*Col2*]),),  
  [*Row1*], "cell-0-0", "cell-1-0",  
  [*Row2*], "cell-0-1", "cell-1-1",  
)
```

```
table(  
  columns: 3,  
  align: center + horizon,  
  stroke: (x:none),  
  table.hline(y:0, stroke:none), //  
remove first line  
  table.hline(y:3, stroke:none), //  
remove last line  
  table.vline(x:1),  
  table.vline(x:2, start:1, end:2,  
stroke: red),  
  table.header([], [*Col1*] ,  
[*Col2*]),),  
  [*Row1*], "cell-0-0", "cell-1-0",  
  [*Row2*], "cell-0-1", "cell-1-1",  
)
```

```
#table(  
  columns: 3,  
  align: center + horizon,  
  table.vline(x:0, stroke: none), table.vline(x:1 , stroke: blue), table.vline(x:2),  
  table.header([], [*Col1*] , [*Col2*], table.hline(stroke: red)),  
  [*Row1*], "cell-0-0", "cell-1-0", table.hline(),  
  [*Row2*], "cell-0-1", "cell-1-1",  
)
```

<i>c</i>	<i>b</i>	<i>a</i>	<i>cb</i>	<i>ba</i>	<i>y</i>
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	0	0
0	1	1	0	1	0
1	0	0	0	0	0
1	0	1	0	0	1
1	1	0	1	0	1
1	1	1	1	1	1

```
#table(  
  columns: 6,  
  stroke: none,  
  align: center+ horizon,  
  table.vline(x:1, stroke:0.5pt), table.vline(x:2, stroke:0.5pt), table.vline(x:3),
```

```

table.vline(x:4, stroke:0.5pt), table.vline(x:5, stroke:0.5pt),
  table.header([$c$], [$b$], [$a$], [$c b$], [$b a$], [$y$], table.hline(stroke: 1pt)),
  [`0`], [`0`], [`0`], [`0`],    [`0`],    [`0`], table.hline(stroke: 0.5pt),
  [`0`], [`0`], [`1`], [`0`],    [`0`],    [`1`], table.hline(stroke: 0.5pt),
  [`0`], [`1`], [`0`], [`0`],    [`0`],    [`0`], table.hline(stroke: 0.5pt),
  [`0`], [`1`], [`1`], [`0`],    [`1`],    [`0`], table.hline(stroke: 1pt),
  [`1`], [`0`], [`0`], [`0`],    [`0`],    [`0`], table.hline(stroke: 0.5pt),
  [`1`], [`0`], [`1`], [`0`],    [`0`],    [`1`], table.hline(stroke: 0.5pt),
  [`1`], [`1`], [`0`], [`1`],    [`0`],    [`1`], table.hline(stroke: 0.5pt),
  [`1`], [`1`], [`1`], [`1`],    [`1`],    [`1`],
)

```

4.7 Icon Boxes

```
#import "/00-templates/boxes.typ": *
```



```
#infobox()["infobox"]
```



```
#ideabox()["ideabox"]
```



```
#warningbox()["warningbox"]
```



```
#importantbox()["importantbox"]
```



```
#firebox()["firebox"]
```



```
#rocketbox()["rocketbox"]
```



```
#todobox()["todobox"]
```



```
#iconbox(icon: "/04-resources/placeholder.svg", linecolor:
hei-blue)["iconbox"]
```

```
#iconbox(linecolor: hei-pink)["iconbox without icon"]
```

4.8 Color Boxes

```
#import "/00-templates/boxes.typ": *
```

Exercise

Some text

```
#colorbox(title: "Exercise", color: hei-blue)[Some text]
```

Attention

Some text

```
#colorbox(title: "Attention", color: hei-pink)[Some text]
```

Consider

Some text

```
#slantedColorbox(title: "Consider", color: hei-green)[Some text]
```

Information

Some text

```
#slantedColorbox(title: "Information", color: hei-orange)[Some text]
```

4.8.1 Todo Box

```
#import "/00-templates/boxes.typ": *
```

TODO

This is not finished

```
#todo("This is not finished")
```

4.8.2 Option Style

The option style allows to underlight a text depending on the type or state of the document. Within the `/01-settings/metadata.typ` the `option.type` can be set to **draft** or **final** or other types.

```
#import "/00-templates/boxes.typ": *
```

```
#option_style(type:"draft")["This text has an option style and is shown in the case the type is draft"]
```

```
#option_style(type:"final")["This text has an option style and is shown in the case the type is final"]
```

« This text has an option style and is shown in the case the type is minimal »

```
#option_style(type:"minimal")["This text has an option style and is shown in the case the type is minimal"]
```

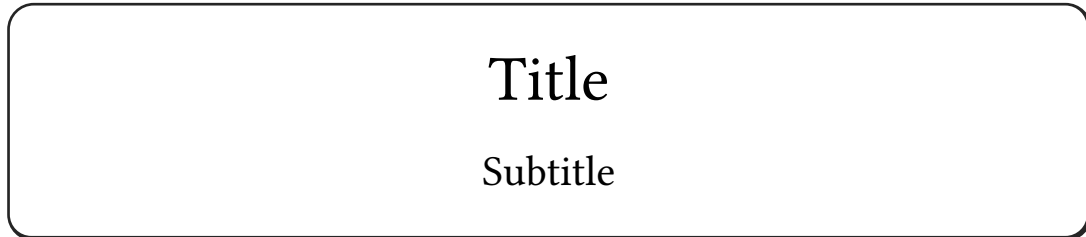
```
#option_style(type:"full")["This text has an option style and is shown in the case the type is full"]
```

```
#option_style(type:"student")["This text has an option style and is shown in the case the type is student"]
```

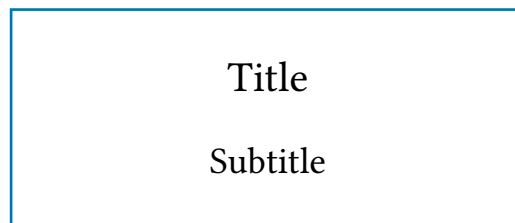
```
#option_style(type:"solution")["This text has an option style and is shown in the case the type is solution"]
```

4.9 Title Box

```
#import "/00-templates/sections.typ": *
```



```
#titlebox(title: [Title], subtitle: [Subtitle])
```



```
#titlebox(width: 50%, radius: 0pt, border: 1pt, linecolor: hei-blue, titlesize: larger, subtitle-size: large, title: [Title], subtitle: [Subtitle])
```



```
#titlebox(linecolor: hei-green, titlesize: larger, subtitle-size: large, title: [Title])
```

4.10 Word/Character Count

With the plugin wordometer is possible to count the words and characters of a text.

```
#import "@preview/wordometer:0.1.4": word-count
```

The following section contains 50 Words and 295 Characters without counting this text.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Quam ipsum accusantium sit nunc nisl, eget lacus vestibulum posuere pellentesque elit. Vivamus magna sed egestas aliquam. Quam nunc sed ipsum nunc atque venenatis hendrerit sed vulgare odio. Nunc sed ipsum nunc atque venenatis hendrerit sed vulgare odio. Nunc sed ipsum nunc atque venenatis hendrerit sed vulgare odio.

```
#word-count(total => [  
  #[The following section contains #total.words Words and #total.characters Characters  
  without counting this text.]<no-wc>  
  #lorem(50)  
], exclude: <no-wc>)
```



The character count does not include spaces.

5 | References

5.1 Links

Example	Raw
https://tschinz.github.io	<code>https://tschinz.github.io</code>
https://tschinz.github.io	<code>#link("https://tschinz.github.io")</code>
See https://tschinz.github.io	<code>#link("https://tschinz.github.io") [See https://tschinz.github.io]</code>
silvan.zahno@hevs.ch	<code>#link("mailto:silvan.zahno@hevs.com") [silvan.zahno@hevs.ch]</code>
	<code>#link("https://tschinz.github.io") [#image(icon, width:0.5cm)]</code>

Table 4 - Links

5.2 Crossreferences

In the document the following references were added.

```
= References <sec:ref>
== Links <sec:links>
#figure(image("/04-resources/icon.svg", width: 2cm)) <fig:icon>
#figure(table(...), kind:table) <tab:links>
#figure(align(left, raw(...)) <code-ref>
$ sum_{k=1}^n k = (n(n+1)) / 2 $ <math-eq1> #ref(<math-eq1>)
```

Listing 1 - Label inserts

They can be references as follows:

Type	Example	Raw
Section	Chapitre 5	<code>@sec:ref</code>
Subsection	Chapitre 5.1	<code>@sec:links</code>
Figure	Figure 1	<code>@fig:icon</code>
Table	Table 4	<code>@tab:links</code>
Code	Listing 1	<code>@code-ref</code>
Equation	Équation 1	<code>@math-eq1</code>

5.3 External References

Example	Raw
[1]	<code>#cite(label("zahnoDynamicProjectPlanning2023"))</code>
[1, p.7ff]	<code>#cite(<zahnoDynamicProjectPlanning2023>, supplement:[p.7ff])</code>
[1]	<code>@zahnoDynamicProjectPlanning2023</code>

5.4 Glossary

The glossary entries need to be defined in `/03-tail/glossary.typ`.

```
#let entry-list = (
  (
    key   : "hei",
    short : "HEI",
    long  : "Haute École d'Ingénierie",
  ),
  (
    key       : "fpga",
    short     : "FPGA",
    plural    : "FPGAs",
    long      : "Field Programmable Gate Array",
    longplural : "Field Programmable Gate Arrays",
    group     : "Technology"
    description : "A field-programmable gate array (FPGA) is an integrated circuit
designed to be configured by a customer or a designer after manufacturing – hence the
term 'field-programmable'.",
  ),
)
```

For the glossary functions the « import » of `/00-templates/helpers.typ` is needed. Underneath it uses the glossarium plugin. It can also be used for acronyms.

```
#import "/00-templates/helpers.typ": *
```

Example

Raw

Haute École d'Ingénierie (HEI) and a second time HEI

```
#gls("hei") and a second time
#gls("hei")
```

Haute École d'Ingénierie (HEI)

```
#gls("hei", long:true)
```

whatever you want

```
#gls("hei", display: "whatever you
want"))
```

Field Programmable Gate Arrays (FPGAs) and a second time FPGAs

```
#glspl("fpga") and a second time
#glspl("fpga")
```

Field Programmable Gate Arrays (FPGAs)

```
#glspl("fpga", long: true))
```

Infotronics (IT) is a specialization of Systems Engineering (SYND) which is part of the HEI. The second time a glossary entry is used the short form will be used: see IT and SYND. To get the long form back use Systems Engineering (SYND).

6 | Code

inline monospaced string

```
`inline monospaced string`
```

```
fn main() {println!("Hello world!")}
```

```
raw(lang:"rust",
  "fn main() {println!(\"Hello world!
  \")"
)
```

```
-- Test 2: INPUT sX, pp
opCode <= "INPUT sX, pp    ";
code    <= "00010";
cIn     <= '0';
A       <= "11110000";
B       <= "00001111";
wait for clockPeriod;
assert Y = "00001111"
  report "test 2 INPUT wrong"
  severity note
```

```
raw(block:true, lang:"vhdl",
  read("code-example.vhdl"))
)
```

```
fn main() {
  println!("Hello world!")
}
```

```
```\rust
fn main() {
 println!("Hello world!")
}
```\
```

```
fn main() {
  println!("Hello world!")
}
```

Listing 2 - Rust Code

```
#figure(
  align(left,
    ```\rust
 fn main() {
 println!("Hello world!")
 }
    ```\
  ),
  caption: [Rust Code],
)
```

A plugin allows to get linenumbers

```
#import "@preview/codelst:2.0.2": sourcecode
```

```
1 fn main() {
2   println!("Hello world!")
3 }
```

```
#sourcecode()[
  ```\rust
 fn main() {
```

```
println!("Hello world!")
}
` `]
```

# 7 | Math Equations

## Inline math

Let  $a$  and  $b$ , and  $c$  be the side of a right-angled triangle.

Let  $a$  and  $b$ , and  $c$  be the side of a right-angled triangle.

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$\sum_{k=1}^n k = (n(n+1)) / 2$ ,

## Fullline math

$$a^2 + b^2 = c^2 \tag{1}$$

$a^2 + b^2 = c^2$  `<math-eq1>`

## Math with caption

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \tag{2}$$

Équation 8 - Some proof

```
#figure(
 $ sum_(k=1)^n k = (n(n+1)) / 2 $,
 caption: [Some proof]
)
```

## 7.1 Align

### Formula

$$\begin{aligned} a_1 &= b_1 + c_1 = z_1 \\ a_2 &= b_2 + c_2 - d_2 + e_2 = z_1 \end{aligned} \tag{3}$$

### Raw

```
$
a_1 = b_1 + c_1 = z_+ \
a_2 = b_2 + c_2 - d_2 + e_2 = z_1
$
```

$$\begin{aligned} a_1 &= b_1 + c_1 &= z_1 \\ a_2 &= b_2 + c_2 - d_2 + e_2 = z_1 \end{aligned} \tag{4}$$

```
$
a_1 &= b_1 + c_1 &= z_1 \
a_2 &= b_2 + c_2 - d_2 + e_2 &= z_1
$
```

## 7.2 Symbols

This is an incomplete list for all symbols goto [here](#)

Outside of the  $\$$  math environment the symboils can be accessed with `.`

## 7.2.1 Accents

Sym-bol	Raw	Sym-bol	Raw	Sym-bol	Raw
$\hat{x}$	<code>\$grave(x)\$</code>	$\acute{x}$	<code>\$acute(x)\$</code>	$\hat{x}$	<code>\$hat(x)\$</code>
$\tilde{x}$	<code>\$tilde(x)\$</code>	$\breve{x}$	<code>\$breve(x)\$</code>	$\dot{x}$	<code>\$dot(x)\$</code>
$\ddot{x}$	<code>\$dot.double(x)\$</code>	$\ddot{x}$	<code>\$dot.triple(x)\$</code>	$\ddot{x}$	<code>\$dot.quad(x)\$</code>
$\ddot{x}$	<code>\$diaer(x)\$</code>	$\mathring{x}$	<code>\$circle(x)\$</code>	$\ddot{x}$	<code>\$acute.double(x)\$</code>
$\tilde{x}$	<code>\$caron(x)\$</code>	$\vec{x}$	<code>\$arrow(x)\$</code>	$\vec{x}$	<code>\$arrow.l(x)\$</code>
$\cancel{x}$	<code>\$cancel(x)\$</code>	$\bar{x}$	<code>\$macron(x)\$</code>	$\overline{xyz}$	<code>\$overline(xyz)\$</code>
$\overline{xyz}$	<code>\$overline(xyz)\$</code>	$\underbrace{xyz}$	<code>\$underbrace(xyz)\$</code>	$\overbrace{xyz}$	<code>\$overbrace(xyz)\$</code>
$\underbrace{xyz}$	<code>\$underbracket(xyz)\$</code>	$\overbrace{xyz}$	<code>\$overbracket(xyz)\$</code>	$\overbrace{xyz}$	<code>\$overbracket(xyz)\$</code>

## 7.2.2 Equals &amp; Operators

Sym-bol	Raw	Sym-bol	Raw	Sym-bol	Raw
$=$	<code>\$=\$</code>	$=$	<code>\$eq\$</code>	$\neq$	<code>\$eq.not\$</code>
$\neq$	<code>\$!= \$</code>	$\equiv$	<code>\$equiv\$</code>	$\neq$	<code>\$equiv.not\$</code>
$\simeq$	<code>\$tilde.eq\$</code>	$\not\equiv$	<code>\$tilde.eq.not\$</code>	$=$	<code>\$eq.small\$</code>
$\geq$	<code>\$gt.eq\$</code>	$\nless$	<code>\$gt.eq.not\$</code>	$\leq$	<code>\$lt.eq\$</code>
$\nless$	<code>\$lt.eq.not\$</code>	$\approx$	<code>\$approx\$</code>	$\approx$	<code>\$approx.eq\$</code>
$\napprox$	<code>\$approx.not\$</code>	$:$	<code>\$colon\$</code>	$:=$	<code>\$colon.eq\$</code>
$=:$	<code>\$eq.colon\$</code>	$::=$	<code>\$colon.double.eq\$</code>	$+$	<code>\$+\$</code>
$+$	<code>\$plus\$</code>	$+$	<code>\$plus.small\$</code>	$\pm$	<code>\$plus.minus\$</code>
$\oplus$	<code>\$plus.circle\$</code>	$-$	<code>\$-\$</code>	$-$	<code>\$minus\$</code>
$\mp$	<code>\$minus.plus\$</code>	$\ominus$	<code>\$minus.circle\$</code>		

## 7.2.3 Scripts

Sym-bol	Raw	Sym-bol	Raw	Sym-bol	Raw
$x_1$	<code>\$x_1\$</code>	$x_{12}$	<code>\$x_(12)\$</code>	$x_1$	<code>\$scripts(x)_1\$</code>
$x_1$	<code>\$x_1\$</code>	$x_{12}$	<code>\$x_(12)\$</code>	$x_1$	<code>\$scripts(x)_1\$</code>
$x_1^2$	<code>\$x_1^2\$</code>	$x_{12}^{34}$	<code>\$x_(12)^(34)\$</code>	$x_1^2$	<code>\$scripts(x)_1^2\$</code>
$x_1^2$	<code>\$x_1^2\$</code>	$x_{12}^{34}$	<code>\$x_(12)^(34)\$</code>	$x_1^2$	<code>\$scripts(x)_1^2\$</code>

## 7.2.4 Special Elements

Symbol	Raw	Symbol	Raw
--------	-----	--------	-----

$$\binom{n}{k} \quad (5) \quad \$ \text{binom}(n, k) \$$$

$$\left\lfloor \frac{1}{2} \right\rfloor \quad (7) \quad \$ \text{round}(1, 2) \$$$

$$\begin{pmatrix} 1 & 2 & \dots & 10 \\ 2 & 2 & \dots & 10 \\ \vdots & \vdots & \ddots & \vdots \\ 10 & 10 & \dots & 10 \end{pmatrix} (9)$$

```
$ mat(
 1, 2, ..., 10;
 2, 2, ..., 10;
 dots.v, dots.v,
 dots.down,
 dots.v;
 10, 10, ...,
 10;
) $
```

$$\sum_{k=0}^n a_k \quad (11) \quad \$ \text{sum\_}(k=0)^n \text{ a\_k} \$$$

$$\sqrt[3]{x} \quad (13) \quad \$ \text{root}(3, x) \$$$

```
$ f(x, y) := cases(
 1 "if" (x dot y)/2 <= 0,
 2 "if" x "is even",
 3 "if" x in NN,
 4 "else",
) $
```

$$\frac{1}{2} \quad (16) \quad \$ \text{frac}(1,2) \$$$

$$\frac{(x+1)}{(x+2)} \quad (18) \quad \$ ((x+1))/(x+2) \$$$

$$n! = \prod_{k=1}^n k \quad (20) \quad \$ \text{product\_}(k=1)^n \text{ k} \$$$

$$\int \quad (22) \quad \$ \text{integral} \$$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad (6) \quad \$ \text{vec}(1, 2, \text{delim:} "[") \$$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad (8) \quad \$ \text{mat}(1,2; 3,4) \$$$

$$\sum a_k \quad (10) \quad \$ \text{sum} \text{ a\_k} \$$$

$$\sum_{k=0}^n a_k \quad (12) \quad \$ \text{scripts}(\text{sum})\_ (k=0)^n \text{ a\_k} \$$$

$$f(x, y) := \begin{cases} 1 & \text{if } \frac{x \cdot y}{2} \leq 0 \\ 2 & \text{if } x \text{ is even} \\ 3 & \text{if } x \in \mathbb{N} \\ 4 & \text{else} \end{cases} \quad (14)$$

$$\frac{1}{2} \quad (15) \quad \$ 1/2 \$$$

$$\frac{x+1}{x+2} \quad (17) \quad \$ (x+1)/(x+2) \$$$

$$\prod \quad (19) \quad \$ \text{product} \$$$

$$n! = \prod_{k=1}^n k \quad (21) \quad \$ \text{n!} = \text{scripts}(\text{product})\_ (k=1)^n \text{ k} \$$$

$$\int_a^b f(x) \quad (23) \quad \$ \text{integral} \$$$

## 7.2.5 Alphabeth

### Symbol

$\alpha\beta\gamma\delta\epsilon\zeta\eta\theta\iota\kappa\lambda\mu\nu\xi\omicron\pi\rho\sigma\tau\upsilon\varphi\chi\psi\omega$

### Raw

$\alpha\beta\gamma\delta\epsilon\zeta\eta\theta\iota\kappa\lambda\mu\nu\xi\omicron\pi\rho\sigma\tau\upsilon\varphi\chi\psi\omega$

ΑΒΓΔΕΖΗΘΙΚΑΜΝΞΟΠΡΣΤΥΦΧΨΩ	omicron pi rho sigma tau upsilon phi chi psi omega\$
ΑΒΓΔΕΖΗΘΙΚΑΜΝΞΟΠΡΣΤΥΦΧΨΩ	\$Alpha Beta Gamma Delta Epsilon Zeta Eta Theta Iota Kappa Lambda Mu Nu Xi Omicron Pi Rho Sigma Tau Upsilon Phi Chi Psi Omega\$
ΑΒΓΔΕΖΗΘΙΚΑΜΝΞΟΠΡΣΤΥΦΧΨΩ	\$AA BB CC DD EE FF GG HH II JJ KK LL MM NN OO PP QQ RR SS TT UU VV WW XX YY ZZ\$

## 7.2.6 Logical

Symbol	Raw	Symbol	Raw	Symbol	Raw
$\wedge$	<code>\$and\$</code>	$\bigwedge$	<code>\$and.big\$</code>	$\&$	<code>\$amp\$</code>
$\vee$	<code>\$or\$</code>	$ $	<code>\$bar.v\$</code>	$*$	<code>\$ast.op\$</code>
$*$	<code>\$ast.basic\$</code>	$*$	<code>\$ast.low\$</code>	$\oplus$	<code>\$plus.circle\$</code>
$\oplus$	<code>\$plus.circle.big\$</code>				

## 7.2.7 Operators

Sym- bol	Raw	Sym- bol	Raw	Sym- bol	Raw
$\sin x$	<code>\$sin x\$</code>	$\cos x$	<code>\$cos x\$</code>	$\tan x$	<code>\$tan x\$</code>
$\arcsin x$	<code>\$arcsin x\$</code>	$\arccos x$	<code>\$arccos x\$</code>	$\arctan x$	<code>\$arctan x\$</code>
$\sinh x$	<code>\$sinh x\$</code>	$\cosh x$	<code>\$cosh x\$</code>	$\tanh x$	<code>\$tanh x\$</code>
$\arg x$	<code>\$arg x\$</code>	$\csc x$	<code>\$csc x\$</code>	$\deg x$	<code>\$deg x\$</code>
$\det x$	<code>\$det x\$</code>	$\dim x$	<code>\$dim x\$</code>	$\exp x$	<code>\$exp x\$</code>
$\bmod x$	<code>\$mod x\$</code>	$\inf x$	<code>\$inf x\$</code>	$\log x$	<code>\$log x\$</code>
$\lim x$	<code>\$lim x\$</code>	$\liminf x$	<code>\$liminf x\$</code>	$\limsup x$	<code>\$limsup x\$</code>
$\min x$	<code>\$min x\$</code>	$\max x$	<code>\$max x\$</code>	$\sup x$	<code>\$sup x\$</code>

## 7.2.8 Arrows

SymRaw	SymRaw	SymRaw	
	Arrows right		
$\rightarrow$	<code>\$arrow\$</code>	$\mapsto$	<code>\$arrow.bar\$</code>
$\mapsto$	<code>\$arrow.bar.long\$</code>	$\Rightarrow$	<code>\$arrow.double.long\$</code>
$\Rightarrow$	<code>\$arrow.double.bar\$</code>	$\Rightarrow$	<code>\$arrow.double.bar.long\$</code>
$\Rightarrow$	<code>\$arrow.stroked\$</code>	$\Rightarrow$	<code>\$arrow.quad\$</code>
$\Rightarrow$	<code>\$arrow.filled\$</code>	$\dashrightarrow$	<code>\$arrow.dashed\$</code>
$\curvearrowright$	<code>\$arrow.curve\$</code>	$\looparrowright$	<code>\$arrow.loop\$</code>
	Arrows left		
$\leftarrow$	<code>\$arrow.l\$</code>	$\leftarrow$	<code>\$arrow.l.bar\$</code>
$\leftarrow$	<code>\$arrow.l.bar.long\$</code>	$\Leftarrow$	<code>\$arrow.l.double\$</code>
		$\Leftarrow$	<code>\$arrow.l.double.long\$</code>

	<code>\$arrow.l.double.bar\$</code>		<code>\$arrow.l.double.bar.long\$</code>		<code>\$arrow.l.quad\$</code>
	<code>\$arrow.l.stroked\$</code>		<code>\$arrow.l.filled\$</code>		<code>\$arrow.l.dashed\$</code>
	<code>\$arrow.l.curve\$</code>		<code>\$arrow.l.squiggly\$</code>		<code>\$arrow.l.loop\$</code>
<b>Double Arrows Left Right</b>					
	<code>\$arrow.l.r\$</code>		<code>\$arrow.l.r.not\$</code>		<code>\$arrow.l.r.long\$</code>
	<code>\$arrow.l.r.double\$</code>		<code>\$arrow.l.r.double.long\$</code>		<code>\$arrow.l.r.double.not\$</code>
	<code>\$arrow.l.r.stroked\$</code>		<code>\$arrow.l.r.filled\$</code>		<code>\$arrow.l.r.wave\$</code>
<b>Arrows Top</b>					
	<code>\$arrow.t\$</code>		<code>\$arrow.t.bar\$</code>		<code>\$arrow.t.double\$</code>
	<code>\$arrow.t.triple\$</code>		<code>\$arrow.t.quad\$</code>		<code>\$arrow.t.stroked\$</code>
	<code>\$arrow.t.filled\$</code>		<code>\$arrow.t.dashed\$</code>		<code>\$arrow.t.curve\$</code>
<b>Arrows Bottom</b>					
	<code>\$arrow.b\$</code>		<code>\$arrow.b.bar\$</code>		<code>\$arrow.b.double\$</code>
	<code>\$arrow.b.triple\$</code>		<code>\$arrow.b.quad\$</code>		<code>\$arrow.b.stroked\$</code>
	<code>\$arrow.b.filled\$</code>		<code>\$arrow.b.dashed\$</code>		<code>\$arrow.b.curve\$</code>
<b>Double Arrows Top Bottom</b>					
	<code>\$arrow.t.b\$</code>		<code>\$arrow.t.b.double\$</code>		<code>\$arrow.t.b.stroked\$</code>
	<code>\$arrow.t.b.filled\$</code>				
<b>Arrows Diagonal Top Right</b>					
	<code>\$arrow.tr\$</code>		<code>\$arrow.tr.double\$</code>		<code>\$arrow.tr.stroked\$</code>
	<code>\$arrow.tr.filled\$</code>		<code>\$arrow.tr.hook\$</code>		
<b>Arrows Diagonal Bottom Right</b>					
	<code>\$arrow.br\$</code>		<code>\$arrow.br.double\$</code>		<code>\$arrow.br.stroked\$</code>
	<code>\$arrow.br.filled\$</code>		<code>\$arrow.br.hook\$</code>		
<b>Arrows Diagonal Bottom Left</b>					
	<code>\$arrow.bl\$</code>		<code>\$arrow.bl.double\$</code>		<code>\$arrow.bl.stroked\$</code>
	<code>\$arrow.bl.filled\$</code>		<code>\$arrow.bl.hook\$</code>		
<b>Arrows Diagonal Top Left</b>					
	<code>\$arrow.tl\$</code>		<code>\$arrow.tl.double\$</code>		<code>\$arrow.tl.stroked\$</code>
	<code>\$arrow.tl.filled\$</code>		<code>\$arrow.tl.hook\$</code>		
<b>Double Arrows Diagonal</b>					
	<code>\$arrow.tl.br\$</code>		<code>\$arrow.tr.bl\$</code>		
<b>Other Arrows</b>					
	<code>\$arrow.cw\$</code>		<code>\$arrow.cw.half\$</code>		<code>\$arrow.ccw\$</code>
	<code>\$arrow.ccw.half\$</code>				

7.2.9 Angles

Sym- bol	Raw	Sym- bol	Raw	Sym- bol	Raw
$\angle$	<code>\$angle\$</code>	$\sphericalangle$	<code>\$angle.rev\$</code>	$\sphericalangle$	<code>\$angle.acute\$</code>
$\sphericalangle$	<code>\$angle.acute\$</code>	$\sphericalangle$	<code>\$angle.arc\$</code>	$\sphericalangle$	<code>\$angle.arc.rev\$</code>
$\sphericalangle$	<code>\$angle.l\$</code>	$\sphericalangle$	<code>\$angle.r\$</code>	$\sphericalangle$	<code>\$angle.l.double\$</code>
$\sphericalangle$	<code>\$angle.r.double\$</code>	$\sphericalangle$	<code>\$angle.right\$</code>	$\sphericalangle$	<code>\$angle.right.rev\$</code>
$\sphericalangle$	<code>\$angle.right.arc\$</code>	$\sphericalangle$	<code>\$angle.right.dot\$</code>	$\sphericalangle$	<code>\$angle.right.sq\$</code>
$\sphericalangle$	<code>\$angle.spheric\$</code>	$\sphericalangle$	<code>\$angle.spheric.rev\$</code>	$\sphericalangle$	<code>\$angle.spheric.top\$</code>

7.2.10 Cool Symbols

Sym- bol	Raw	Sym- bol	Raw	Sym- bol	Raw
@	<code>\$at\$</code>	%	<code>\$co\$</code>	©	<code>\$copyright\$</code>
©	<code>\$copyright.sound\$</code>	°C	<code>\$degree.c\$</code>	€	<code>\$euro\$</code>
\$	<code>\$dollar\$</code>	£	<code>\$pound\$</code>	¥	<code>\$won\$</code>
¥	<code>\$yen\$</code>	฿	<code>\$bitcoin\$</code>	°F	<code>\$degree.f\$</code>
!	<code>\$excl\$</code>	!	<code>\$excl.inv\$</code>	!!	<code>\$excl.double\$</code>
!?	<code>\$excl.quest\$</code>	↯	<code>\$arrow.zigzag\$</code>	⊗	<code>\$ast.circle\$</code>
✱	<code>\$ast.triple\$</code>	χ	<code>\$chi\$</code>	☐	<code>\$floral\$</code>
⌘	<code>\$maltese\$</code>	¶	<code>\$pilcrow\$</code>	ℏ	<code>\$planck\$</code>
♣	<code>\$suit.club\$</code>	♦	<code>\$suit.diamond\$</code>	♥	<code>\$suit.heart\$</code>
♠	<code>\$suit.spade\$</code>	△	<code>\$triangle.stroked.nested2\$</code>		

7.2.11 Style

Symbol	Raw	Symbol	Raw
ABC123	<code>\$sans(A B C 1 2 3)\$</code>	ℳℳℳ123	<code>\$frak(A B C 1 2 3)\$</code>
ABC123	<code>\$mono(A B C 1 2 3)\$</code>	ABC123	<code>\$bb(A B C 1 2 3)\$</code>
ABC123	<code>\$cal(A B C 1 2 3)\$</code>		

Symbol	Raw
--------	-----

$\sum_{i \in \mathbb{N}} 1 + i$

```
#show math.equation: set text(font: "Fira Math")
$sum_(i in NN) 1 + i$,
```



## 8 | Emoji Symbols

This is an incomplete list for all emoji goto [here](#)

If the emoji module is imported the `#emoji` can be removed

```
#import emoji: *
```

Sym Raw

Sym Raw

Sym Raw

Smileys

	<code>#emoji.face</code>		<code>#emoji.baby</code>		<code>#emoji.alien</code>
	<code>#emoji.alien.monster</code>		<code>#emoji.alien.monster</code>		<code>#emoji.eye</code>
	<code>#emoji.eyes</code>		<code>#emoji.face.grin</code>		<code>#emoji.face.angry</code>
	<code>#emoji.face.angry.red</code>		<code>#emoji.face.anguish</code>		<code>#emoji.face.astonish</code>
	<code>#emoji.face.bandage</code>		<code>#emoji.face.beam</code>		<code>#emoji.face.concern</code>
	<code>#emoji.face.cool</code>		<code>#emoji.face.cover</code>		<code>#emoji.face.down</code>
	<code>#emoji.face.down.sweat</code>		<code>#emoji.face.drool</code>		<code>#emoji.face.explode</code>
	<code>#emoji.face.eyeroll</code>		<code>#emoji.face.friendly</code>		<code>#emoji.face.fear</code>
	<code>#emoji.face.fear.sweat</code>		<code>#emoji.face.fever</code>		<code>#emoji.face.flush</code>
	<code>#emoji.face.halo</code>		<code>#emoji.face.happy</code>		<code>#emoji.face.heart</code>
	<code>#emoji.face.hearts</code>		<code>#emoji.face.heat</code>		<code>#emoji.face.kiss</code>
	<code>#emoji.face.kiss.heart</code>		<code>#emoji.face.kiss.blush</code>		<code>#emoji.face.joy</code>
	<code>#emoji.face.lie</code>		<code>#emoji.face.mask</code>		<code>#emoji.face.meh</code>
	<code>#emoji.face.melt</code>		<code>#emoji.face.neutral</code>		<code>#emoji.face.party</code>
	<code>#emoji.face.peek</code>		<code>#emoji.face.plead</code>		<code>#emoji.face.relief</code>
	<code>#emoji.face.vomit</code>		<code>#emoji.face.unhappy</code>		<code>#emoji.face.wow</code>
	<code>#emoji.face.yawn</code>		<code>#emoji.face.wink</code>		<code>#emoji.face.zip</code>
	<code>#emoji.finger.l</code>		<code>#emoji.finger.t</code>		<code>#emoji.finger.b</code>
	<code>#emoji.finger.m</code>		<code>#emoji.finger.front</code>		<code>#emoji.fingers.cross</code>
	<code>#emoji.fingers.pinch</code>		<code>#emoji.fingers.snap</code>		<code>#emoji.fist.front</code>
	<code>#emoji.fist.r</code>		<code>#emoji.fist.l</code>		<code>#emoji.fist.raised</code>
	<code>#emoji.glasses</code>		<code>#emoji.glasses.sun</code>		<code>#emoji.hand.raised</code>
	<code>#emoji.hand.r</code>		<code>#emoji.hand.l</code>		<code>#emoji.hand.t</code>

## Sym Raw

 #emoji.hand.b  
 #emoji.hand.love  
 #emoji.hand.pinch  
 #emoji.hand.write  
 #emoji.hands.heart

## Sym Raw

 #emoji.hand.ok  
 #emoji.hand.part  
 #emoji.hand.rock  
 #emoji.hands.folded  
 #emoji.hands.shake

## Sym Raw

 #emoji.hand.call  
 #emoji.hand.peace  
 #emoji.hand.wave  
 #emoji.hands.clap  
 #emoji.hand.splay

## Drinks

## Sym Raw

 #emoji.beer  
 #emoji.coffee  
 #emoji.glass.clink  
 #emoji.teacup

## Sym Raw

 #emoji.beer.clink  
 #emoji.cup.straw  
 #emoji.glass.milk  
 #emoji.mate

## Sym Raw

 #emoji.cocktail.tropical  
 #emoji.glass.tumbler  
 #emoji.teapot  
 #emoji.soup

## Animals

## Sym Raw

 #emoji.ant  
 #emoji.beetle  
 #emoji.beetle.lady  
 #emoji.bison  
 #emoji.butterfly  
 #emoji.cat  
 #emoji.chipmunk  
 #emoji.crocodile  
 #emoji.dodo  
 #emoji.duck  
 #emoji.fairy  
 #emoji.flamingo  
 #emoji.giraffe

## Sym Raw

 #emoji.badger  
 #emoji.bee  
 #emoji.beaver  
 #emoji.boar  
 #emoji.camel  
 #emoji.chicken  
 #emoji.cockroach  
 #emoji.dino.pod  
 #emoji.dog  
 #emoji.egg  
 #emoji.feather  
 #emoji.fly  
 #emoji.hedgehog

## Sym Raw

 #emoji.bat  
 #emoji.beetle  
 #emoji.bear  
 #emoji.bug  
 #emoji.camel.dromedar  
 #emoji.chicken.male  
 #emoji.cricket  
 #emoji.dino.rex  
 #emoji.dolphin  
 #emoji.elephant  
 #emoji.fish  
 #emoji.goat  
 #emoji.hippo

## Flowers

## Sym Raw

 #emoji.fleur  
 #emoji.flower.pink

## Sym Raw

 #emoji.flower.hibiscus  
 #emoji.flower.rose

## Sym Raw

 #emoji.flower.lotus  
 #emoji.flower.sun

## Sym Raw

 `#emoji.flower.wilted`

## Sym Raw

 `#emoji.flower.yellow`

## Sym Raw

 `#emoji.plant`

## Food

## Sym Raw

 `#emoji.avocado` `#emoji.broccoli` `#emoji.carrot` `#emoji.cucumber`

## Sym Raw

 `#emoji.aubergine` `#emoji.burger` `#emoji.chocolate` `#emoji.fries`

## Sym Raw

 `#emoji.banana` `#emoji.cake.slice` `#emoji.corn` `#emoji.honey`

## Objects

## Sym Raw

 `#emoji.bin` `#emoji.books` `#emoji.bell` `#emoji.broom` `#emoji.cabinet.file` `#emoji.compass`

## Sym Raw

 `#emoji.bomb` `#emoji.book` `#emoji.bell.not` `#emoji.brush` `#emoji.chair` `#emoji.comet`

## Sym Raw

 `#emoji.bone` `#emoji.book.open` `#emoji.briefcase` `#emoji.bubbles` `#emoji.coffin` `#emoji.hook`

## Sport

## Sym Raw

 `#emoji.baseball` `#emoji.chess`

## Sym Raw

 `#emoji.billiards` `#emoji.climbing`

## Sym Raw

 `#emoji.boxing` `#emoji.football`

## Vehicles

## Sym Raw

 `#emoji.airplane` `#emoji.boat.sail` `#emoji.car` `#emoji.ship`

## Sym Raw

 `#emoji.bike` `#emoji.boat.speed` `#emoji.car.front` `#emoji.motorcycle`

## Sym Raw

 `#emoji.boat` `#emoji.cablecar` `#emoji.car.racing` `#emoji.wheelchair.motor`

## Building

## Sym Raw

 `#emoji.castle.eu` `#emoji.factory` `#emoji.house`

## Sym Raw

 `#emoji.castle.jp` `#emoji.hospital` `#emoji.museum`

## Sym Raw

 `#emoji.circus` `#emoji.office` `#emoji.crane`

## Weather

## Sym Raw

## Sym Raw

## Sym Raw

## Sym Raw

	<code>#emoji.cloud</code>
	<code>#emoji.cloud.snow</code>
	<code>#emoji.cloud.sun.hidden</code>
	<code>#emoji.drop</code>

## Sym Raw

	<code>#emoji.cloud.dust</code>
	<code>#emoji.cloud.storm</code>
	<code>#emoji.cloud.sun.rain</code>
	<code>#emoji.drops</code>

## Sym Raw

	<code>#emoji.cloud.rain</code>
	<code>#emoji.cloud.sun</code>
	<code>#emoji.cloud.thunder</code>
	<code>#emoji.sun</code>

## Special

## Sym Raw

✓	<code>#emoji.checkmark.heavy</code>
	<code>#emoji.circle.white</code>
✗	<code>#emoji.crossmark</code>
	<code>#emoji.explosion</code>
	<code>#emoji.flag.white</code>
	<code>#emoji.flag.red</code>
	<code>#emoji.globe.eu.af</code>
	<code>#emoji.notes</code>

## Sym Raw

	<code>#emoji.checkmark.box</code>
	<code>#emoji.circle.stroked</code>
	<code>#emoji.crossmark.box</code>
	<code>#emoji.fire</code>
	<code>#emoji.flag.black</code>
	<code>#emoji.globe.am</code>
	<code>#emoji.globe.meridian</code>
	<code>#emoji.notes.triple</code>

## Sym Raw

	<code>#emoji.circle.black</code>
©	<code>#emoji.copyright</code>
!	<code>#emoji.excl</code>
	<code>#emoji.firecracker</code>
	<code>#emoji.flag.goal</code>
	<code>#emoji.globe.as.au</code>
#	<code>#emoji.hash</code>
*	<code>#emoji.ast</code>

## Technology

## Sym Raw

	<code>#emoji.battery</code>
	<code>#emoji.brightness.low</code>
	<code>#emoji.calendar</code>
	<code>#emoji.cassette</code>
	<code>#emoji.clamp</code>
	<code>#emoji.clock.alarm</code>
	<code>#emoji.controller</code>
	<code>#emoji.email</code>
	<code>#emoji.folder</code>
	<code>#emoji.hammer</code>
	<code>#emoji.laptop</code>

## Sym Raw

	<code>#emoji.battery.low</code>
	<code>#emoji.bubble.speech.l</code>
	<code>#emoji.camera</code>
	<code>#emoji.clip</code>
	<code>#emoji.clock.one</code>
	<code>#emoji.computer</code>
	<code>#emoji.crab</code>
	<code>#emoji.floppy</code>
	<code>#emoji.folder.open</code>
	<code>#emoji.hammer.wrench</code>

## Sym Raw

	<code>#emoji.brightness.high</code>
	<code>#emoji.bubble.speech.r</code>
	<code>#emoji.camera.movie</code>
	<code>#emoji.clipboard</code>
	<code>#emoji.clock.two</code>
	<code>#emoji.computermouse</code>
	<code>#emoji.disc.cd</code>
	<code>#emoji.flashlight</code>
	<code>#emoji.gear</code>
	<code>#emoji.headphone</code>

# Glossaire

**FPGA** – Field Programmable Gate Array: A field-programmable gate array (FPGA) is an integrated circuit designed to be configured by a customer or a designer after manufacturing – hence the term 'field-programmable'. [24](#)

**HEI** – Haute École d'Ingénierie [24](#)

**IT** – Infotronics [24](#)

**SYND** – Systems Engineering [24](#)

# Bibliographie

- [1] S. Zahno *et al.*, « Dynamic Project Planning with Digital Twin », *Frontiers in Manufacturing Technology*, vol. 3, mai 2023, doi: [10.3389/fmtec.2023.1009633](https://doi.org/10.3389/fmtec.2023.1009633).